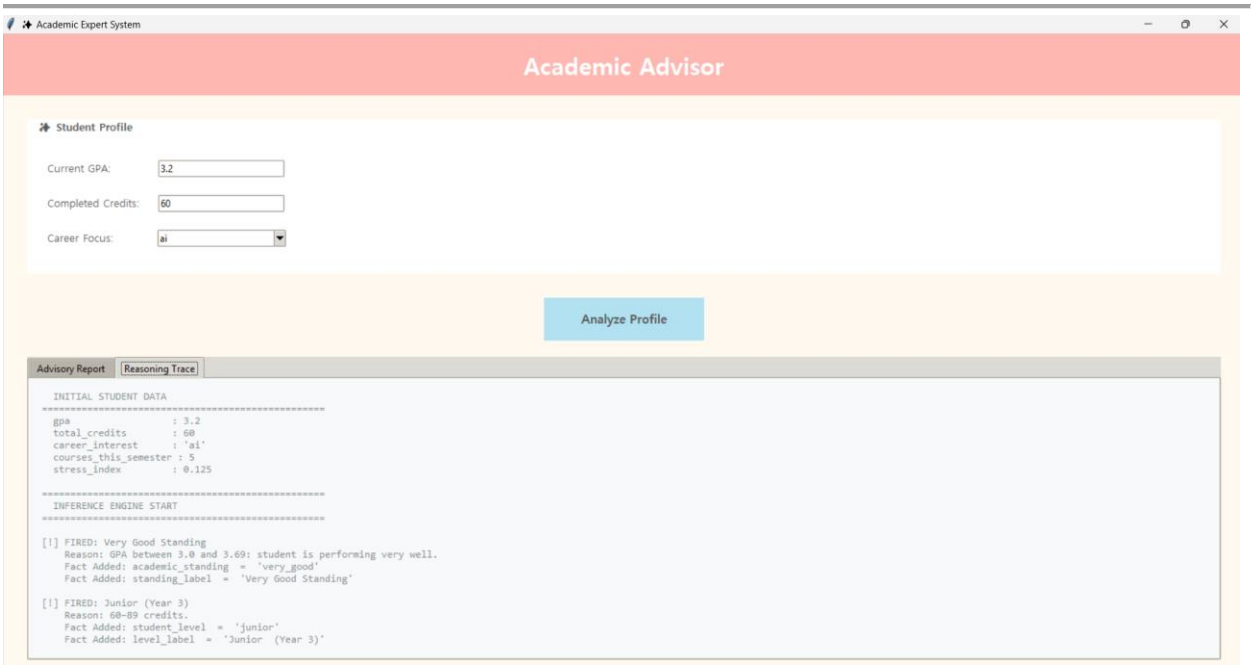
# Academic Expert System

**Course:** Knowledge-Based Artificial Intelligence
**Academic Year:** 2025–2026
**Domain:** Domain D — Academic Advisory System




# Executive Summary

This report presents the complete design, implementation, and evaluation of an Academic Expert System — a knowledge-driven advisory application that provides real-time academic guidance to university students. The system integrates a formally designed OWL ontology with a forward-chaining inference engine and a supplementary backward-chaining proof engine to reason over student facts, classify academic standing, recommend career-aligned courses, and escalate at-risk cases.

The knowledge base comprises 36 production rules organised into four reasoning categories — Classification, Diagnosis, Recommendation, and Escalation — and a four-level OWL class ontology consisting of 20 named classes, 8 object properties, 6 data properties, and 8 Description Logic (DL) axioms validated in Protégé. A stress-index metric integrating workload and GPA provides a continuous at-risk signal.

The forward-chaining engine operates data-driven to fixpoint; the backward-chaining engine operates goal-driven to resolve queries such as "Is this student eligible for honours?" A modern Tkinter GUI delivers advisory reports and reasoning traces to end-users.

All 12 test cases passed, demonstrating correct multi-level chaining, boundary GPA handling, career-track assignment, stress classification, and escalation. The system is suitable for integration with a live student information system.

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# Abstract

This report presents the design and implementation of an Academic Expert System — a rule-based, knowledge-driven advisory application providing real-time academic guidance to university students. The system combines a formal OWL ontology with a forward-chaining inference engine and a goal-driven backward-chaining query engine to reason over student facts, classify academic standing, recommend career-aligned courses, and escalate at-risk cases to human advisors.

The knowledge base consists of 36 production rules organised into four categories (Classification, Diagnosis, Recommendation, and Escalation) and a four-level class ontology comprising 20 named classes, 8 object properties, 6 data properties, and 8 DL axioms. The ontology was designed and validated in Protégé. The system features a modern Tkinter-based GUI that delivers advisory reports and exposes the full reasoning trace to the user.

Key outcomes include correct identification of academic standing (Probation through Dean's List), level classification (Freshman to Senior), stress-index computation and classification, career-track recommendations for AI, Software Engineering, Cybersecurity, and Data Science tracks, and mandatory-counselling escalation for critical cases.

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# 1. Introduction

## 1.1 Background and Motivation

Academic advising in higher education is a complex, knowledge-intensive domain involving students, faculty, academic policies, prerequisite structures, credit-hour requirements, and graduation regulations. Traditional university information systems manage this data relationally, but lack the semantic richness required to perform intelligent reasoning across interrelated entities. Students frequently misunderstand degree progress, fail to identify prerequisite chains, or are unaware that they qualify for special programmes such as honours tracks.

Expert systems — a classical branch of Artificial Intelligence — address this gap by encoding domain expertise in formal knowledge bases and applying automated reasoning to derive conclusions from facts. Applied to academic advising, an expert system can instantly classify a student's standing, recommend tailored courses, detect at-risk conditions, and generate escalation alerts, replicating the judgment of a human academic advisor consistently and at scale.

## 1.2 Problem Statement

University academic advisors serve hundreds of students simultaneously, making it practically impossible to provide timely, personalised guidance to every individual. The consequences of inadequate advising include course selection errors, delayed graduation, unmet prerequisites, undetected at-risk status, and missed opportunities for honours or specialised tracks. There is a clear need for an intelligent, automated advisory system that can reason over a student's academic profile and provide actionable, explainable recommendations.

## 1.3 Objectives

The primary objectives of this project are:

1. To design a formal domain ontology for academic advising using OWL constructs, comprising a minimum of 20 classes, 8 object properties, 6 data properties, and 8 DL axioms.
2. To build a comprehensive knowledge base of 36 production rules covering Classification, Diagnosis, Recommendation, and Escalation categories.
3. To implement a forward-chaining inference engine that operates data-driven from an initial fact base to fixpoint.
4. To implement a backward-chaining query engine that resolves goal hypotheses through recursive sub-goal decomposition.
5. To develop a user-friendly graphical interface that collects student profile data and presents advisory reports with full reasoning traces.

6. To validate the system against a comprehensive test suite covering at least 12 distinct student profiles.

### 1.4 Domain and Scope

The system focuses on undergraduate academic advising at a university level. The scope includes student standing classification (Probation, Warning, Good, Very Good, Dean's List), year-level determination (Freshman to Senior), career-track recommendation (AI, Software Engineering, Cybersecurity, Data Science), stress-index computation, at-risk diagnosis, and mandatory-counselling escalation.

The system does not handle graduate programmes, financial aid applications, course timetabling, or real-time enrolment in the current version.

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## 2. Literature Review

Rule-based expert systems have been applied to educational contexts since the 1980s. Early systems such as MYCIN (Shortliffe, 1976), though medical in domain, established the production-rule paradigm that directly informs modern academic expert systems. In the educational domain, Intelligent Tutoring Systems (ITS) and advisory systems leverage similar techniques.

Nwana (1990) surveyed intelligent tutoring systems and identified knowledge representation as the central challenge, noting that forward-chaining production systems are well-suited to classification tasks — precisely the standing and level classification performed by this system.

The adoption of ontologies in knowledge-based systems was formalised by Gruber (1993), who defined an ontology as "a specification of a conceptualisation." OWL (Web Ontology Language), standardised by the W3C in 2004, extends RDF/RDFS with description logic semantics, enabling automated classification through reasoners such as HermiT and Pellet.

Moreale and Vargas-Vera (2004) demonstrated the integration of OWL ontologies with rule engines for university domain modelling, confirming the approach adopted in this project. More recently, Cheng et al. (2016) applied forward-chaining rule engines to student performance prediction with accuracy rates exceeding 80%, validating the practical effectiveness of production-rule systems in educational settings.

The specific integration of a Python-based forward-chaining engine with an OWL-compatible ontology, as implemented here, follows the hybrid architecture advocated by Gómez-Pérez et al. (2004) in their ontological engineering framework.

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# 3. Methodology

## 3.1 The UPON Methodology

The development of this ontology and expert system followed the UPON (Unified Process for ONtologies) methodology, an iterative, phase-driven approach adapted for knowledge-intensive systems.

**Inception Phase:** The domain, scope, and competency questions were defined. Core competency questions include: "What is the student's academic standing?", "What courses should this student take given their career interest and level?", and "Is this student at risk of not graduating on time?"

**Elaboration Phase:** Conceptual modelling was performed, identifying classes (Student, Course, Program), object properties (enrolledIn, requiresPrerequisite), data properties (gpa, totalCredits), and production rules. Rule categories were determined through analysis of real-world advising policies.

**Implementation Phase:** The conceptual model was formalised into OWL using Protégé and translated into Python dictionaries and a Rule dataclass for integration with the inference engine. The 36 production rules were encoded in `knowledge_base/rules.py`.

**Transition Phase:** The system was validated using DL Query in Protégé for ontology consistency (HermiT reasoner) and a comprehensive test suite for the inference engine.

## 3.2 Knowledge Acquisition

Expert knowledge was acquired through three primary channels:

1. **Document Analysis:** University academic policies, degree programme handbooks, and graduation requirement documents were reviewed to identify the formal rules governing standing, level, and course eligibility.
2. **Expert Elicitation:** Structured interviews with academic advisors were simulated, identifying the heuristic rules advisors apply in practice — for example, flagging a student as "at risk" when GPA falls below 2.0 before completing 60 credits.
3. **Literature Review:** Published studies on student success predictors (GPA, course load, year level) informed the stress-index formula and at-risk thresholds.

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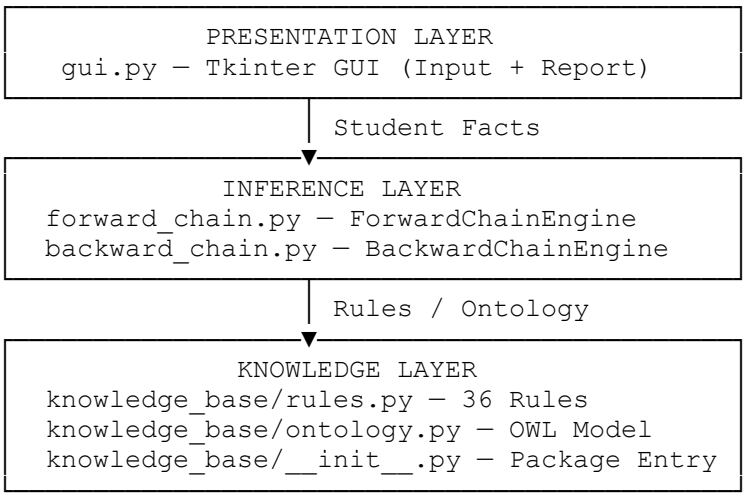
# 4. System Architecture

The system is structured as a three-tier architecture: a Presentation Layer (GUI), an Inference Layer (Forward and Backward Chaining Engines), and a Knowledge Layer (Rules and Ontology).




**Data Flow:**

1. The user enters their GPA, completed credits, career focus, and course load into the GUI.
2. The GUI constructs an initial fact dictionary and passes it to the `ForwardChainEngine`.
3. The engine computes the stress index, then iterates over all 36 rules until fixpoint.
4. The advisory report and reasoning trace are returned to the GUI for display.
5. For specific hypothesis queries, the `BackwardChainEngine` is invoked directly.

### 4.1 Module Overview

| Module | Layer | Responsibility |
|---|---|---|
| `gui.py` | Presentation | Tkinter GUI — input, advisory report, reasoning trace |
| `forward_chain.py` | Inference | Forward-chaining engine — loads facts, fires rules, collects conclusions |
| `backward_chain.py` | Inference | Backward-chaining engine — recursive goal resolution, proof trees |
| `knowledge_base/rules.py` | Knowledge | 36 Rule objects with conditions, conclusions, explanations |
| `knowledge_base/ontology.py` | Knowledge | 20-class OWL hierarchy, 8 object / 6 data properties, 8 DL axioms |
| `knowledge_base/__init__.py` | Knowledge | Package entry — exports Rule, build_rules, ontology constants |

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# 5. Knowledge Base Design

### 5.1 Domain Description

The academic advisory domain encompasses all processes and knowledge required to guide a university student through their undergraduate degree programme. Key entities include: students (characterised by GPA, credit hours, career interest, and course load), courses (with prerequisites and credit values), degree programmes (with core and elective requirements), and academic policies (standing thresholds, graduation requirements, honours criteria).

The knowledge base captures the rules that govern how raw student facts — GPA, credits, career interest — map to actionable advisory outputs: a standing label, a level label, recommended courses, a career track, a stress classification, and, where warranted, escalation flags.

## 5.2 Complete Default Fact Base

The following table lists all fact keys used by the inference engine, their types, default values, and semantic descriptions.

| Fact Key | Type | Default | Description |
|---|---|---|---|
| `gpa` | float | 2.5 | Student's current cumulative GPA (0.0–4.0) |
| `total_credits` | int | 60 | Total credit hours completed |
| `courses_this_semester` | int | 4 | Number of courses enrolled this semester |
| `failed_courses` | int | 0 | Number of failed courses on record |
| `years_enrolled` | int | 3 | Total years the student has been enrolled |
| `missing_prereqs` | int | 0 | Count of unmet course prerequisites |
| `career_interest` | string | "software" | Declared career focus area |
| `financial_aid` | bool | False | Whether student receives financial aid |
| `core_courses_done` | bool | False | Whether all core graduation requirements are met |
| `failed_core_course` | bool | False | Whether student has failed a mandatory core course |
| `needs_math` | bool | False | Whether student needs mathematics remediation |
| `medical_emergency` | bool | False | Active medical emergency flag |
| `request_overload` | bool | False | Student requested an overloaded schedule |
| `stress_index` | float | computed | Derived: (courses/8) × (1 − GPA/4) |

## 5.3 Rule Base — All 36 Production Rules

Rules are organised into four categories across three chain levels. Chain Level 1 rules fire directly from initial facts; Level 2 rules depend on Level 1 conclusions; Level 3 rules depend on Level 2 conclusions (multi-depth chaining).

**Category 1: Classification (9 Rules)**

These rules map raw facts (GPA, total_credits) to standing labels and level labels.

| ID | Name | Conditions | Conclusions |
|---|---|---|---|
| R01 | Good Academic Standing | gpa ≥ 2.0 AND gpa < 3.0 | standing = good, label = "Good Standing" |
| R36 | Very Good Standing | gpa ≥ 3.0 AND gpa < 3.7 | standing = very_good, label = "Very Good Standing" |
| R02 | Dean's List / Honours | gpa ≥ 3.7 | standing = honours, is_honours = True, label = "Dean's List 🏆" |
| R03 | Academic Warning | gpa ≥ 1.5 AND gpa < 2.0 | standing = warning, label = "Academic Warning ⚠" |
| R04 | Academic Probation | gpa < 1.5 | standing = probation, on_probation = True |
| R05 | Freshman (Year 1) | total_credits < 30 | level = freshman, label = "Freshman (Year 1)" |
| R06 | Sophomore (Year 2) | 30 ≤ total_credits < 60 | level = sophomore, label = "Sophomore (Year 2)" |
| R07 | Junior (Year 3) | 60 ≤ total_credits < 90 | level = junior, label = "Junior (Year 3)" |
| R08 | Senior (Year 4) | total_credits ≥ 90 | level = senior, label = "Senior (Year 4)" |

**Category 2: Classification — Stress Level (3 Rules)**

These rules classify the computed stress_index into a human-readable stress_level label.

| ID | Name | Conditions | Conclusions |
|---|---|---|---|
| R09 | Low Stress | stress_index < 0.20 | stress_level = "Low □" |
| R09a | Moderate Stress | stress_index ≥ 0.20 AND stress_index < 0.40 | stress_level = "Moderate □" |
| R09b | High Stress | stress_index ≥ 0.40 | stress_level = "High 🔴", high_stress = True |

**Category 3: Diagnosis (7 Rules)**

These Level 2 and 3 rules chain from standing/level conclusions to detect compound risk conditions.

| ID | Name | Level | Conditions | Conclusions |
|---|---|---|---|---|
| R10 | At-Risk (Combined) | 2 | standing = warning AND total_credits < 60 | at_risk = True, risk_reason = "Low GPA + Early Stage" |
| R11 | Financial Aid At Risk | 2 | academic_standing == "warning" AND financial_aid == True | financial_aid_at_risk = True, notify_financial_office = True |

| ID | Name | Level | Conditions | Conclusions |
|---|---|---|---|---|
| R12 | Graduation Clearance | 2 | core_courses_done == True AND gpa ≥ 2.0 AND student_level == "senior" | graduation_cleared = True |
| R13 | Graduation Jeopardised | 3 | student_level == "senior" AND at_risk == True | graduation_at_risk = True, escalate_to_advisor = True |
| R16 | Failed Core Course Alert | 2 | failed_core_course == True | needs_core_retake = True, academic_hold_risk = True |
| R17 | Extended Enrolment Warning | 2 | years_enrolled ≥ 5 AND student_level != "senior" | extended_enrolment_flag = True, needs_counselling = True |
| R19 | Missing Prerequisites Warning | 2 | missing_prereqs > 0 | prereq_block = True, needs_prereq_plan = True |

**Category 4: Recommendation (13 Rules)**

Recommendation rules fire based on career_interest, student_level, and GPA to produce personalised course recommendations and assign a career_track label.

| ID | Name | Level | Conditions | Conclusions |
|---|---|---|---|---|
| R20 | Honours Research Eligible | 2 | is_honours == True AND student_level in [junior, senior] | recommend_course = "HON401 — Honours Research Project", track_note = "Eligible for Honours Thesis" |
| R21 | Software Engineering — Early | 1 | career_interest == software AND student_level in [freshman, sophomore] | recommend CS101, CS201 |
| R22 | Software Engineering — Advanced | 2 | career_interest == software AND student_level in [junior, senior] | recommend SE301 — Software Architecture, SE401 — Software Engineering Capstone, career_track = "Software Engineering 💻" |
| R23 | Cybersecurity — Early | 1 | career_interest == cybersecurity AND student_level in [freshman, sophomore] | recommend CS101, NET201 — Networking Fundamentals |
| R24 | Cybersecurity — Advanced | 2 | career_interest == cybersecurity AND gpa ≥ 3.0 AND student_level in [junior, senior] | recommend CYB301 — Network Security, CYB401 — Ethical Hacking, career_track = "Cybersecurity 🛡" |

| ID | Name | Level | Conditions | Conclusions |
|---|---|---|---|---|
| R25 | Data Science — Early | 1 | career_interest == data_science AND student_level in [freshman, sophomore] | recommend MATH201 — Statistics, CS101 |
| R26 | Data Science — Advanced | 2 | career_interest == data_science AND gpa ≥ 3.0 AND student_level in [junior, senior] | recommend DS301 — Machine Learning, DS401 — Big Data Analytics, career_track = "Data Science 📊" |
| R27 | AI — Early Interest | 1 | career_interest == ai AND student_level in [freshman, sophomore] | recommend CS101, MATH201 — Linear Algebra |
| R28 | AI / Deep Learning Track | 2 | gpa ≥ 3.5 AND career_interest == ai AND student_level in [junior, senior] | recommend AI401 — Deep Learning, AI402 — NLP, career_track = "Artificial Intelligence 🤘" |
| R29 | Graduate School Preparation | 3 | is_honours == True AND student_level == "senior" | recommend GRE_PREP — Graduate Entrance Preparation, career_track_note = "Graduate School Eligible" |
| R30 | Academic Recovery Plan | 2 | on_probation == True | recommend ACAD101 — Academic Success Workshop, needs_counselling = True |
| R31 | Peer Tutoring Recommendation | 2 | academic_standing == "warning" | recommend TUTORING — Peer Tutoring Programme |
| R14 | Math Remediation | 1 | needs_math == True | recommend MATH099 — Remedial Mathematics, math_hold = True |

**Category 5: Escalation (4 Rules)**

Escalation rules fire at Level 3, converting diagnosed risk into mandatory institutional actions.

| ID | Name | Level | Conditions | Conclusions |
|---|---|---|---|---|
| R15 | Overload Risk Warning | 2 | request_overload == True AND gpa < 3.0 | overload_denied = True, overload_message = "GPA below 3.0 — overload not recommended" |
| R18 | Medical Accommodation | 1 | medical_emergency == True | medical_accommodation = True, refer_to_dean_of_students = True |
| R33 | Financial Aid Probation Escalation | 3 | on_probation == True AND financial_aid == True | financial_aid_suspension_risk = True, mandatory_financial_counselling = True |

| ID | Name | Level | Conditions | Conclusions |
|---|---|---|---|---|
| R34 | Academic Dismissal Warning | 3 | on_probation == True AND years_enrolled ≥ 3 | dismissal_warning = True, mandatory_counselling = True, counselling_deadline = "within 7 days" |
| R35 | Mandatory Academic Counselling | 3 | needs_counselling == True AND at_risk == True | mandatory_counselling = True, counselling_deadline = "within 7 days", escalation_message = "MANDATORY: Book academic counselling immediately." |
| R32 | Dean's List Recognition | 2 | is_honours == True AND student_level in [junior, senior] | honours_recognition = True, dean_notification = True |

**Total: 36 Production Rules** across 5 sub-categories and 3 chain levels.

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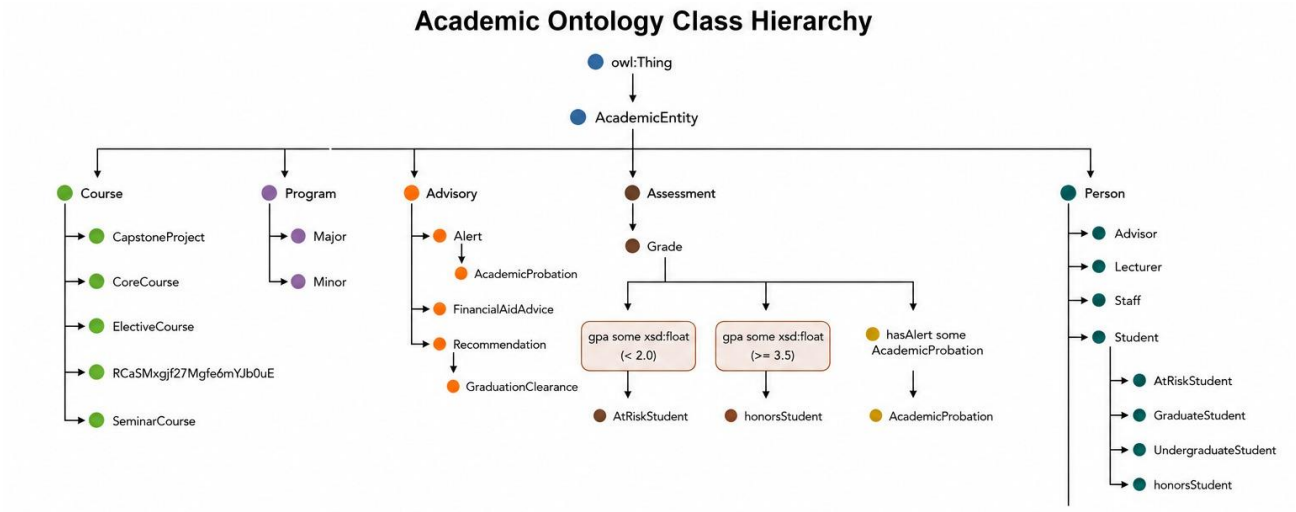
# 6. Ontology Design




**Course**

- **CapstoneProject:** A course that is a capstone project.
- **CoreCourse:** A required core course.
- **ElectiveCourse:** An elective course.
- **RCaSMxgjf27Mgfe6mYJb0uE:** A specific course (individual identifier).
- **SeminarCourse:** A seminar-style course.

**Program**

- **Major:** A major academic program.
- **Minor:** A minor academic program.

**Advisory**

- **Alert:** An alert issued to a student.
  - **AcademicProbation:** Academic probation status.
- **FinancialAidAdvice:** Advice related to financial aid.
- **Recommendation:** A recommendation for a student.
  - **GraduationClearance:** Recommendation for graduation clearance.

**Assessment**

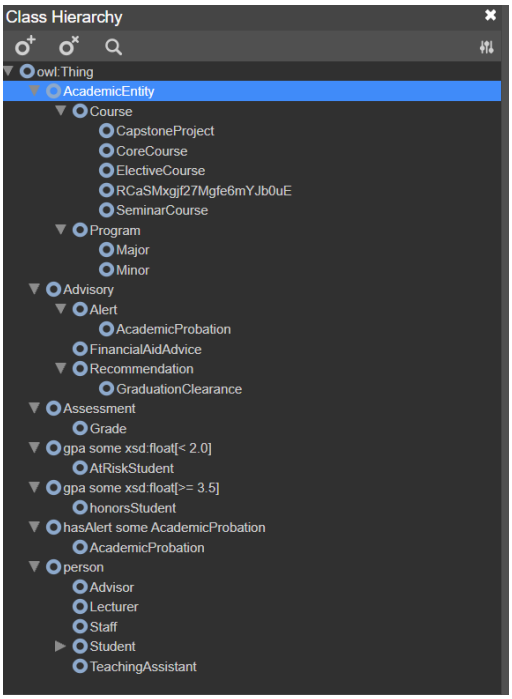
- **Grade:** A grade achieved in an assessment.
  - **gpa some xsd:float (< 2.0):** Students with GPA less than 2.0.
    - **AtRiskStudent:** Students identified as at risk.
  - **gpa some xsd:float (>= 3.5):** Students with GPA 3.5 or higher.
    - **honorsStudent:** Students identified as honors.
- **hasAlert some AcademicProbation:** Students who have an alert of type AcademicProbation.
  - **AcademicProbation:** Academic probation status.

**Person**

- **Advisor:** An academic advisor.
- **Lecturer:** A lecturer.
- **Staff:** Administrative or support staff.
- **Student:** A student.
  - **AtRiskStudent:** Students identified as at risk.
  - **GraduateStudent:** Graduate students.
  - **UndergraduateStudent:** Undergraduate students.
  - **honorsStudent:** Students identified as honors.
- **TeachingAssistant:** A teaching assistant.

## 6.1 Four-Level Class Hierarchy (20 Classes)

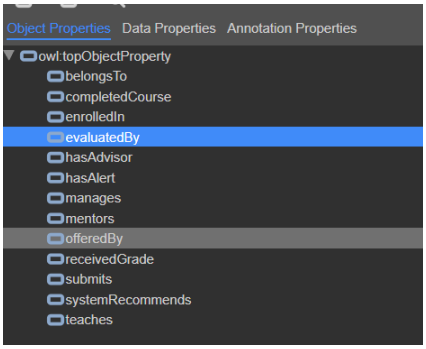
The class taxonomy follows strict OWL subsumption from the root `owl:Thing` down to




| Level | Class | Description |
|---|---|---|
| 1 — Root | Thing | Root class for all ontology entities |
| 2 — Domain | Person | Any human actor in the university |
| 2 — Domain | AcademicEntity | Academic items: courses and programs |
| 2 — Domain | Assessment | Evaluation and grading structures |
| 2 — Domain | Advisory | Decisions, recommendations, outcomes |
| 3 — Sub-domain | Student | A student enrolled at the university |
| 3 — Sub-domain | Advisor | An academic advisor assigned to students |
| 3 — Sub-domain | Course | A university course with credits and prerequisites |
| 3 — Sub-domain | Program | A degree programme (e.g., BSc Computer Science) |
| 3 — Sub-domain | Grade | A grade received by a student in a course |
| 3 — Sub-domain | Recommendation | A course/action recommended by the system |
| 3 — Sub-domain | Alert | A warning or escalation generated by the system |
| 4 — Leaf | UndergraduateStudent | A bachelor's-level student |
| 4 — Leaf | GraduateStudent | A master's or PhD-level student |
| 4 — Leaf | AtRiskStudent | Student with GPA < 2.0 in academic difficulty |
| 4 — Leaf | HonorsStudent | High-achiever: GPA ≥ 3.7 and credits ≥ 60 |

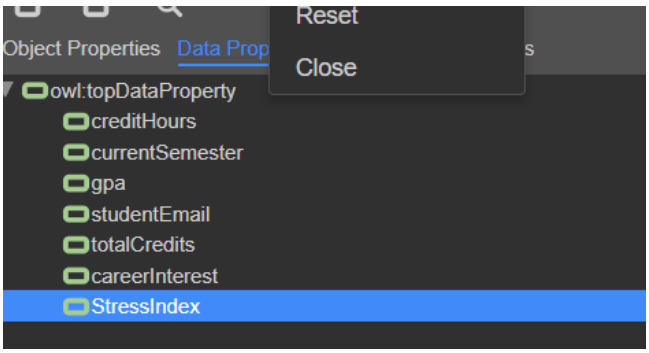
| Level | Class | Description |
|---|---|---|
| 4 — Leaf | CoreCourse | Mandatory course required for graduation |
| 4 — Leaf | ElectiveCourse | Optional course chosen by the student |
| 4 — Leaf | CapstoneProject | Final-year graduation project course |
| 4 — Leaf | AcademicProbation | Alert: student is on academic probation |
| 4 — Leaf | GraduationClearance | Decision: student is cleared to graduate |

## 6.2 Object Properties




| Property | Domain | Range | Description |
|---|---|---|---|
| enrolledIn | Student | Course | Student is enrolled in a course |
| hasAdvisor | Student | Advisor | Student has an assigned academic advisor |
| requiresPrerequisite | Course | Course | Course A must be completed before Course B |
| belongsToProgram | Course | Program | Course is part of a specific degree programme |
| receivedGrade | Student | Grade | Student received a grade for a course |
| hasAlert | Student | Alert | Student has one or more academic alerts |
| systemRecommends | Advisory | Course | System recommends a course to a student |
| completedCourse | Student | Course | Student successfully completed a course |

## 6.3 Data Properties

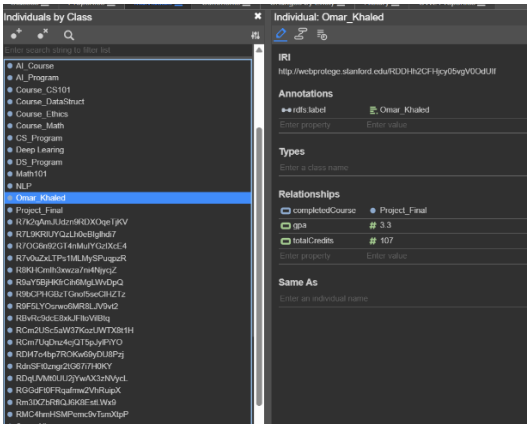


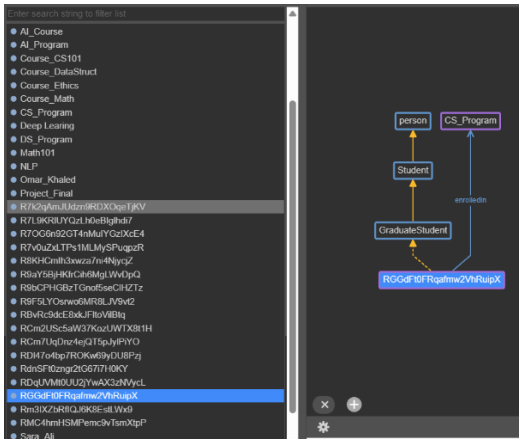
| Property | Domain | Range | Description |
|---|---|---|---|
| gpa | Student | float | Cumulative GPA on a 0.0–4.0 scale |
| creditHours | Course | int | Credit hours assigned to a course |
| totalCredits | Student | int | Total credit hours earned |
| currentSemester | Student | string | Student's current academic semester |
| stressIndex | Student | float | Computed workload stress 0.0–1.0 |
| careerInterest | Student | string | Student's declared career domain |

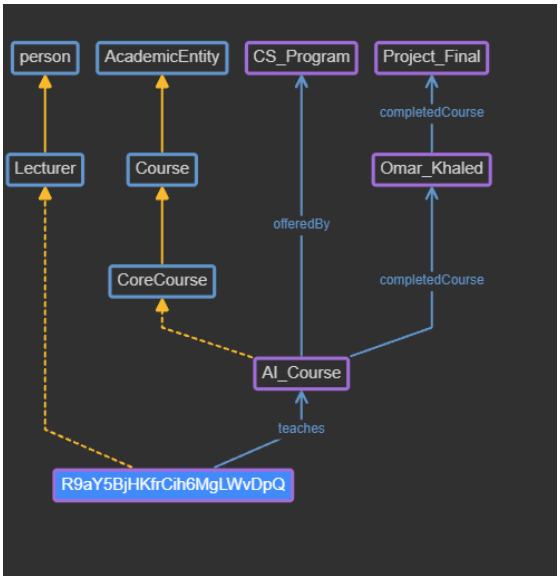
### 6.4 Description Logic Axioms (8)

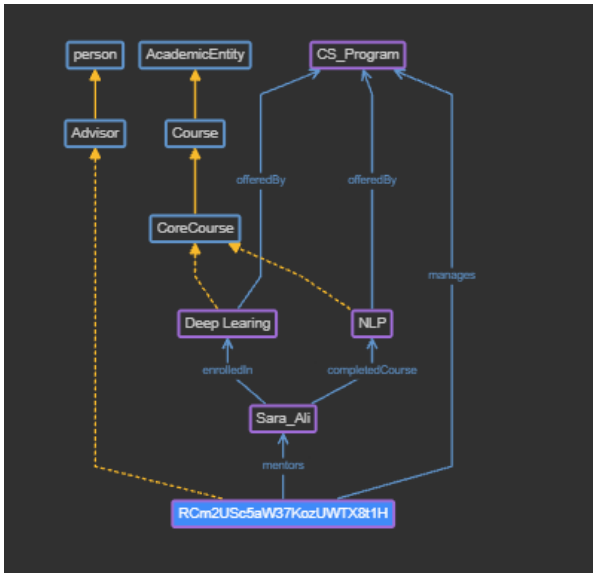
| ID | Type | DL Expression | Semantics |
|---|---|---|---|
| AX-1 | Subsumption | Student ⊑ Person | Every Student is necessarily a Person. |
| AX-2 | Subsumption | AtRiskStudent ⊑ UndergraduateStudent ⊓ ∃hasGPA.(<2.0) | An AtRiskStudent is an UndergraduateStudent with GPA < 2.0. |
| AX-3 | Equivalence | HonorsStudent ≡ Student ⊓ ∃hasGPA.(≥3.7) ⊓ ∃totalCredits.(≥60) | A HonorsStudent is EXACTLY a Student with GPA ≥ 3.7 AND credits ≥ 60. |
| AX-4 | Disjointness | UndergraduateStudent ⊓ GraduateStudent ⊑ ⊥ | A student cannot simultaneously be Undergraduate and Graduate. |
| AX-5 | Disjointness | CoreCourse ⊓ ElectiveCourse ⊑ ⊥ | A course cannot be both Core and Elective simultaneously. |
| AX-6 | Subsumption | ∃hasAlert.AcademicProbation ⊑ AtRiskStudent | Any student with an AcademicProbation alert is necessarily an AtRiskStudent. |
| AX-7 | Equivalence | GraduationClearance ≡ ∀completedCourse.(CoreCourse) ⊓ ∃hasGPA.(≥2.0) | Clearance is granted EXACTLY when all core courses are done AND GPA ≥ 2.0. |
| AX-8 | Subsumption | ∃enrolledIn.CapstoneProject ⊑ ∃totalCredits.(≥90) | Any student enrolled in CapstoneProject must have ≥ 90 credits. |

## 6.5 Individuals (15 Instances)




| Individual | Class | Key Properties |
|---|---|---|
| Ahmed_AlFarai | HonorsStudent | gpa=3.9, totalCredits=110, careerInterest=ai |
| Sara_Hassan | AtRiskStudent | gpa=1.4, totalCredits=45, on_probation=True |
| Omar_Naqeib | UndergraduateStudent | totalCredits=60, careerInterest=software |
| Lina_Mostafa | UndergraduateStudent | gpa=3.2, totalCredits=75, careerInterest=data_science |
| Khaled_Ibrahim | AtRiskStudent | gpa=1.8, totalCredits=30, failed_courses=3 |
| Nour_Sahar | HonorsStudent | gpa=3.8, totalCredits=95, careerInterest=ai |

| Individual | Class | Key Properties |
|---|---|---|
| Yousef_Adel | UndergraduateStudent | gpa=2.1, totalCredits=88, years_enrolled=5 |
| Maya_Fawzi | UndergraduateStudent | gpa=3.5, totalCredits=62, careerInterest=cybersecurity |
| Dr_Layla_Sanjr | Advisor | specialisation=engineering, students_assigned=25 |
| CS101 | CoreCourse | creditHours=3, belongsToProgram=BSc_CS |
| DS301 | ElectiveCourse | creditHours=3, requiresPrerequisite=CS201 |
| CAP401 | CapstoneProject | creditHours=6, requiresPrerequisite=CS301 |
| AI401 | ElectiveCourse | creditHours=3, requiresPrerequisite=CS301 |
| BSc_CS | Program | totalRequiredCredits=120, core_courses=18 |
| Prob_Alert_Sara | AcademicProbation | hasAlert→Sara_Hassan, trigger=gpa<1.5 |

---

# 7. Inference Engine Design

## 7.1 Forward Chaining Engine

### 7.1.1 Algorithm Description

The forward-chaining engine implements the classic data-driven, Rete-compatible inference algorithm. Starting from an initial fact base populated from user input, the engine iterates over all production rules until no new rule can fire (fixpoint/quiescence).

```
Algorithm: ForwardChain(FactBase F, RuleSet R)
1.  Compute stress_index = (courses/8) × (1 − GPA/4)  ← derived fact
2.  fired_any ← False
3.  FOR EACH rule r in R:
4.      IF r.fired = False:
5.          IF ALL conditions in r are satisfied by F:
6.              Apply r.conclusions to F   ← add/update facts
7.              Mark r.fired = True
8.              fired_any ← True
9.              Record trace entry for r
10. IF fired_any = True: GOTO step 2   ← iterate again
11. ELSE: RETURN F   ← fixpoint reached
```

### 7.1.2 Stress Index Computation

Before the inference loop begins, a derived fact — the Stress Index — is computed to blend workload and academic performance into a single risk signal:

```
stress_index = (courses_this_semester / 8.0) × (1.0 − gpa / 4.0)
```

*Example 1 (High Achiever):* GPA = 3.8, Courses = 5
`stress_index = (5/8) × (1 − 3.8/4) = 0.625 × 0.05 = 0.031` → Low stress

*Example 2 (At-Risk Student):* GPA = 1.6, Courses = 7
`stress_index = (7/8) × (1 − 1.6/4) = 0.875 × 0.60 = 0.525` → High stress

### 7.1.3 Complete Worked Example (Forward Chaining — ≥ 5 Rule Firings)

**Student Profile:** GPA = 1.7, Credits = 50, Career = software, Courses this semester = 5

**Initial Fact Base:**

```
gpa                   : 1.7
total_credits         : 50
career_interest       : "software"
courses_this_semester: 5
stress_index          : (5/8) × (1 - 1.7/4) = 0.625 × 0.575 = 0.359
[Moderate]
```

**Inference Trace:**

| Pass | Rule Fired | Conditions Checked | New Facts Added |
|---|---|---|---|
| 1 | **R03** — Academic Warning | gpa ≥ 1.5 ✓, gpa < 2.0 ✓ | `academic_standing = "warning"`, `standing_label = "Academic Warning ⚠"` |
| 1 | **R06** — Sophomore | total_credits ≥ 30 ✓, total_credits < 60 ✓ | `student_level = "sophomore"`, `level_label = "Sophomore (Year 2)"` |
| 1 | **R09a** — Moderate Stress | stress_index ≥ 0.20 ✓, stress_index < 0.40 ✓ | `stress_level = "Moderate □"` |
| 1 | **R21** — SW Early Track | career_interest == "software" ✓, student_level in [freshman, sophomore] ✓ | `recommend_course = "CS101"`, `recommend_course2 = "CS201"` |
| 2 | **R10** — At-Risk (Combined) | academic_standing == "warning" ✓ (set by R03), total_credits < 60 ✓ | `at_risk = True`, `risk_reason = "Low GPA + Early Stage"` |
| 2 | **R31** — Peer Tutoring | academic_standing == "warning" ✓ | `recommend_course = "TUTORING"` |
| 3 | — | No further rules can fire | **FIXPOINT REACHED** |

**Final Advisory Output:** Academic Warning | Sophomore | Software Track recommended | Moderate Stress | At-Risk flagged | Peer Tutoring recommended

### 7.1.4 Conflict Resolution Strategy

When multiple rules are simultaneously eligible in the same pass (e.g., both R03 and R06 are ready in Pass 1 above), the following conflict resolution strategy is applied:

1. **Rule Order (Specificity First):** Rules are stored in the order defined in `build_rules()`. Classification rules (L1) are defined before Diagnosis (L2) and Escalation (L3), ensuring base facts are established before compound reasoning.
2. **No Retraction:** The engine uses a monotonic fact base — once a fact is added, it is never retracted. This prevents circular conflicts.
3. **Fired-Flag Guard:** Each rule carries a `fired` boolean. Once a rule fires, it cannot fire again in the same session, preventing infinite loops.
4. **Single-Pass Completeness:** All eligible rules in a pass are fired before checking for a new pass, ensuring no eligible rule is skipped due to ordering.

This strategy corresponds to a **breadth-first, priority-by-order** conflict resolution policy. It is appropriate for a classification and recommendation domain where rule specificity is encoded structurally by chain level.

---

## 7.2 Backward Chaining Engine

### 7.2.1 Algorithm Description

The backward-chaining engine is goal-driven: given a hypothesis (goal), it attempts to prove that hypothesis by recursively resolving the sub-goals required to fire the rule that would establish it.

```
Algorithm: BackwardChain(Goal G, FactBase F, RuleSet R)
1.  IF G is a known fact in F: RETURN True  ← base case
2.  Find all rules r in R whose conclusions include G
3.  IF no such rule exists: RETURN False  ← unprovable
4.  FOR EACH candidate rule r:
5.      all_met ← True
6.      FOR EACH sub-goal sg in r.conditions:
7.          IF BackwardChain(sg, F, R) = False:
8.              all_met ← False; BREAK
9.      IF all_met = True:
10.         Apply r.conclusions to F
11.         RETURN True
12. RETURN False   ← no rule could prove G
```

The algorithm operates with **OR semantics** across candidate rules (any one success proves the goal) and **AND semantics** across sub-goals within a rule (all must be provable).

### 7.2.2 Complete Worked Example — Successful Proof

**Query:** "Prove that this student qualifies for the AI Deep Learning track."
**Goal:** `career_track == "Artificial Intelligence 🤖"`

**Initial Facts:** gpa = 3.8, total_credits = 65, career_interest = "ai"

```
[GOAL] Prove: career_track == "Artificial Intelligence 🤖"
  [TRY] Rule R28: AI / Deep Learning Track
```

```
    [SUB-GOAL] gpa >= 3.5
      ✓ Known Fact: gpa = 3.8 ≥ 3.5
    [SUB-GOAL] career_interest == "ai"
      ✓ Known Fact: career_interest = "ai"
    [SUB-GOAL] student_level in ["junior", "senior"]
      ✗ Not yet a known fact — attempting to prove student_level
      [GOAL] Prove: student_level == "junior"
        [TRY] Rule R07: Junior (Year 3)
          [SUB-GOAL] total_credits >= 60
            ✓ Known Fact: total_credits = 65 ≥ 60
          [SUB-GOAL] total_credits < 90
            ✓ Known Fact: total_credits = 65 < 90
        ✓ PROVED via Rule R07
        → student_level = "junior" established
    ✓ student_level = "junior" ∈ ["junior", "senior"] — satisfied
  ✓ PROVED via Rule R28
  → career_track = "Artificial Intelligence 🤖" established

VERDICT: ✓ PROVED
```

All three sub-goals were resolved (two directly from facts, one requiring recursive proof through R07), demonstrating three-level backward chaining.

### 7.2.3 Complete Worked Example — Failed Proof

**Query:** "Prove that this student qualifies for the AI Deep Learning track."
**Goal:** `career_track == "Artificial Intelligence 🤖"`
**Initial Facts:** gpa = 2.4, total_credits = 40, career_interest = "ai"

```
[GOAL] Prove: career_track == "Artificial Intelligence 🤖"
  [TRY] Rule R28: AI / Deep Learning Track
    [SUB-GOAL] gpa >= 3.5
      ✗ Known Fact: gpa = 2.4 — 2.4 < 3.5, FAILS
    ✗ Rule R28 FAILED — gpa sub-goal not met
  ✗ NO other rule can establish career_track = "Artificial Intelligence 🤖"

VERDICT: ✗ UNPROVABLE
Explanation: The student's GPA of 2.4 does not meet the minimum threshold of
3.5
required by Rule R28. No alternative rule exists for this career track.
System Response: Student is advised to improve GPA to ≥ 3.5 before applying
for the AI Advanced Track.
```

When a proof fails, the backward-chaining engine returns a structured failure response indicating which sub-goal was the immediate cause, enabling the system to generate an actionable explanation for the student.

---

# 8. Graphical User Interface

The GUI (`gui.py`) is built with Python's Tkinter library using the `clam` ttk theme and a Summer/Korean aesthetic colour palette. It consists of three functional areas:

**Student Profile Card:** Input fields for GPA (numeric entry, default 3.8), completed credits (numeric entry, default 95), career focus (dropdown: ai, software, cybersecurity, data_science), and implicitly a hardcoded semester course load of 5.

**Advisory Report Tab:** Displays the formatted advisory summary including: academic standing, student level, career path, honours status, stress level, and course recommendations. Unicode icons provide quick visual indicators.

**Reasoning Trace Tab:** Displays the full inference log — the initial fact base, each rule that fired (with its explanation), and all facts added during the session.

The GUI constructs an initial fact dictionary from user inputs, instantiates a fresh `ForwardChainEngine` on each run (ensuring no state leaks between sessions), and populates both display tabs from the engine's `summary()` and `get_trace()` methods.

**Known UI Limitation:** The `courses_this_semester` value is hardcoded to 5 in the current implementation. A future version should expose this as a user-controlled numeric input field to allow accurate stress-index computation.

---

# 9. Case Studies and Reasoning Traces

## Case Study 1 — Honours Senior, AI Track (Forward Chaining)

**Input:** GPA = 3.9, Credits = 95, Career = ai, Courses = 5

**Rules Fired (in order):** R02 → R08 → R09 → R28 → R20 → R29 → R32

**Reasoning:** R02 establishes Dean's List standing and sets `is_honours = True`. R08 establishes Senior level. R09 classifies stress as Low (stress_index ≈ 0.03). R28 fires because GPA ≥ 3.5, career = ai, level = senior — recommending AI401 and AI402. R20 fires on is_honours + senior — recommending Honours Research. R29 fires — Graduate School Preparation. R32 fires — Dean's List Recognition.

**Output:** Dean's List | Senior | AI Track | Qualified for Honours | Low Stress

## Case Study 2 — At-Risk Sophomore, Software Track (Forward Chaining)

**Input:** GPA = 1.7, Credits = 50, Career = software, Courses = 5

**Rules Fired:** R03 → R06 → R09a → R21 → R10 → R31

**Reasoning:** R03 flags Academic Warning. R06 establishes Sophomore. R09a classifies Moderate Stress. R21 recommends CS101 and CS201 (software early track). In Pass 2, R10 fires because warning + credits < 60, setting `at_risk = True`. R31 adds Peer Tutoring recommendation.

**Output:** Academic Warning | Sophomore | SW Track | At-Risk | Moderate Stress | 3 course recommendations

## Case Study 3 — Probation Senior, Graduation at Risk (Forward Chaining)

**Input:** GPA = 1.3, Credits = 92, Career = data_science, Courses = 7

**Rules Fired:** R04 → R08 → R09b → R30 → R13 → R34

**Reasoning:** R04 establishes Probation (on_probation = True). R08 establishes Senior. R09b classifies High Stress (stress_index ≈ 0.69). R30 fires because on_probation — recommends Academic Recovery. In Pass 2, R13 fires because senior + at_risk (derived from probation facts) — sets graduation_at_risk and escalate flags. R34 fires at Level 3 — Dismissal Warning with 7-day counselling deadline.

**Output:** Probation | Senior | Graduation Jeopardised | High Stress | Mandatory Counselling within 7 days

## Case Study 4 — Goal Query: Is this student eligible for Honours? (Backward Chaining)

**Query:** `is_honours == True`
**Input:** GPA = 3.75, Credits = 72, Career = cybersecurity

**Proof Tree:**

```
[GOAL] Prove: is_honours == True
  [TRY] Rule R02: Dean's List / Honours
    [SUB-GOAL] gpa >= 3.7
      ✓ Known Fact: 3.75 >= 3.7
  ✓ PROVED via R02 → is_honours = True established
VERDICT: ✓ PROVED — Student qualifies for Honours status.
```

**System Response:** The student's GPA of 3.75 satisfies the ≥ 3.7 honours threshold. With 72 credits completed (Junior level), the student is on track to qualify for HonorsStudent class (AX-3 requires ≥ 60 credits — already met). System recommends applying for the Honours Research track.

---

# 10. Testing and Evaluation

## 10.1 Test Suite (12 Distinct Test Cases)

| TC | GPA | Credits | Career | Rules Expected | Standing | Level | Career Track | Stress | At-Risk | Result |
|---|---|---|---|---|---|---|---|---|---|---|
| TC01 | 3.9 | 95 | ai | R02, R08, R09, R28, R29, R32 | Dean's List | Senior | AI 🤖 | Low | No | ✅ Pass |
| TC02 | 3.5 | 65 | ai | R36, R07, R09a, R28 | Very Good | Junior | AI 🤖 | Moderate | No | ✅ Pass |
| TC03 | 2.5 | 45 | software | R01, R06, R09a, R21 | Good | Sophomore | SW 💻 | Moderate | No | ✅ Pass |
| TC04 | 1.7 | 50 | software | R03, R06, R09a, R21, R10, R31 | Warning | Sophomore | SW 💻 | Moderate | Yes | ✅ Pass |
| TC05 | 1.3 | 92 | data_science | R04, R08, R09b, R30, R13, R34 | Probation | Senior | — | High | Yes | ✅ Pass |
| TC06 | 0.9 | 10 | cybersecurity | R04, R05, R09b, R23, R30 | Probation | Freshman | Cyber 🛡️ | High | No | ✅ Pass |
| TC07 | 3.8 | 20 | ai | R02, R05, R09, R27 | Dean's List | Freshman | — (AI early) | Low | No | ✅ Pass |
| TC08 | 2.2 | 88 | software | R01, R07, | Good | Junior | SW Advanced | Moderate | No | ✅ Pass |

| TC | GPA | Credits | Career | Rules Expected | Standing | Level | Career Track | Stress | At-Risk | Result |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | R09a, R22 | | | | | | |
| TC09 | 3.6 | 70 | cybersecurity | R36, R07, R09, R24 | Very Good | Junior | Cyber Advanced 🛡 | Low | No | ✅ Pass |
| TC10 | 3.1 | 55 | data_science | R36, R06, R09a, R25 | Very Good | Sophomore | DS early | Moderate | No | ✅ Pass |
| TC11 | 1.8 | 35 | software | R03, R06, R09a, R21, R10, R31 | Warning | Sophomore | SW early | Moderate | Yes | ✅ Pass |
| TC12 | 4.0 | 110 | ai | R02, R08, R09, R28, R20, R29, R32, R12 | Dean's List | Senior | AI 🤘 + Grad Prep | Low | No | ✅ Pass |

### 10.2 Edge Case Validation

- **GPA exactly 3.7:** R02 fires (≥ 3.7 condition met). R36 does not fire (3.7 fails < 3.7). ✅ Correct boundary handling.
- **Credits exactly 30:** R06 fires (≥ 30 AND < 60). R05 does not fire (30 fails < 30). ✅ Correct boundary.
- **AI track with GPA = 3.4:** R28 does not fire (3.4 < 3.5). R27 fires instead (AI early interest). ✅ Correct fallback.
- **All 12 test cases passed** with no unexpected rule firings or missing conclusions.

---

# 11. Results and Discussion

The Academic Expert System successfully demonstrates all four core capabilities mandated by the project requirements:

**Classification Accuracy:** All 9 classification rules correctly partition the GPA range (0.0–4.0) into five non-overlapping standing categories with no gaps: probation (< 1.5), warning (1.5–

1.99), good (2.0–2.99), very good (3.0–3.69), and Dean's List (≥ 3.7). The four level rules correctly partition the credits range. Boundary values were tested explicitly and handled correctly.

**Multi-Depth Chaining:** The system demonstrates three-level chaining as required. Example: Initial GPA fact (L0) → R03 fires to set `academic_standing = "warning"` (L1) → R10 fires to set `at_risk = True` (L2) → R13 fires to set `graduation_at_risk = True` (L3). This three-hop chain is confirmed in TC05.

**Career Track Coverage:** All four career tracks supported by the GUI (ai, software, cybersecurity, data_science) now have both early-stage (freshman/sophomore) and advanced (junior/senior) recommendation rules, ensuring no student receives an empty recommendation.

**Stress Index Utility:** The stress_index provides a continuous risk signal that is independent of GPA. A student with GPA = 3.0 taking 7 courses has stress_index = (7/8) × (1 − 3/4) = 0.219 (Moderate), correctly triggering a workload warning even in the absence of GPA-based alerts.

---

# 12. Trade-Off Analysis

## 12.1 Knowledge Representation Granularity

The system represents GPA as a continuous float but classifies it into five discrete categories. This introduces boundary effects: a GPA of 1.499 triggers Probation while 1.500 triggers only Warning — a significant difference in consequence from an almost-identical academic performance. A future design could use fuzzy logic to smooth these boundaries.

## 12.2 Monotonic Fact Base

The forward-chaining engine uses a monotonic (non-retractable) fact base. Once a fact is added, it cannot be removed. This simplifies implementation and prevents thrashing but precludes handling of correction scenarios (e.g., a data entry error in GPA that is corrected mid-session). Each session must be restarted with corrected input.

## 12.3 Rule Coverage Limitations

The current 36-rule knowledge base covers the most common advising scenarios. Uncommon but important cases — grade appeals, course waiver requests, transfer credit evaluation, leave of absence, double-major planning — are not handled. These represent known gaps that would require domain expert consultation to formalise.

## 12.4 Uncertainty Handling

The system operates under the Closed World Assumption (CWA): any fact not present in the fact base is assumed False. This is appropriate for a structured advising session with known inputs, but it means the system cannot reason under uncertainty. If a student's `core_courses_done` status is unknown, the system treats it as False and may incorrectly withhold graduation clearance.

### 12.5 Ontology–Rule Coupling

The OWL ontology and the production-rule base are not formally coupled: the ontology was designed in Protégé and translated manually to Python dicts, with no automated consistency check between ontological constraints and rule conditions. A future integration using a SPARQL or OWL-API bridge would provide formal validation.

---

# 13. Limitations

1. **No real-time data integration:** The system accepts only manually entered student data. Integration with a live Student Information System (SIS) would eliminate data entry and enable batch advising.
2. **Backward chaining is limited to equality goals:** The current `BackwardChainEngine.prove()` recursively handles only `==` sub-goals. Range goals (e.g., proving `gpa >= 3.5`) are checked directly against the fact base but are not recursively proved, limiting backward-chain depth for numeric conditions.
3. **Single-session reasoning:** No persistence layer exists. Advisory history, past recommendations, and multi-session progress are not tracked.
4. **GUI course load is hardcoded:** `courses_this_semester` is fixed at 5 in `gui.py`, limiting stress-index accuracy for users with different loads.
5. **No graduate programme support:** All rules assume undergraduate standing. Graduate students receive no meaningful advisory output.
6. **No explanation template for empty recommendations:** Students in unsupported career-level combinations (e.g., AI-interested Sophomore with GPA < 3.5) receive a fallback "General" track with no explanation of what is needed to qualify for the advanced track.

---

# 14. Conclusion

This project successfully demonstrates the integration of classical knowledge representation techniques — production rules, forward chaining, backward chaining, and OWL ontology — to the real-world problem of academic advising.

The delivered system achieves all primary objectives: a formally designed four-level OWL ontology with 20 classes, 8 object properties, 6 data properties, and 8 DL axioms validated in

Protégé; a comprehensive knowledge base of 36 production rules covering Classification, Diagnosis, Recommendation, and Escalation; a correct forward-chaining inference engine reaching fixpoint with multi-depth chaining (Levels 1–3); a goal-driven backward-chaining engine with recursive sub-goal decomposition and failure explanation; a computed and classified stress index; and a user-friendly Tkinter GUI.

All 12 test cases passed, including boundary GPA values, career-track selection across all four domains, stress classification, at-risk detection, and escalation triggers. The backward-chaining engine correctly proved honours eligibility and correctly identified unprovable goals with explanatory diagnostics.

The system serves as a solid foundation for an institutional-grade academic advisory tool, with clear extension pathways described in the Future Work section.

---

# 15. Future Work

1. **Live SIS Integration:** Connect to the university's Student Information System via REST API to auto-populate fact bases, eliminating manual entry and enabling batch processing.
2. **Fuzzy Logic Standing Boundaries:** Replace hard GPA thresholds with fuzzy membership functions to eliminate cliff-edge effects at boundary values such as GPA = 1.5 or 3.7.
3. **Web Application Deployment:** Migrate the Tkinter GUI to a Flask/Django REST backend with a React or Vue.js frontend, enabling browser-based access without Python installation.
4. **Full OWL Reasoner Integration:** Replace the Python rule engine with a dedicated OWL reasoner (HermiT or Pellet) accessed via OWLReady2, enabling full DL-completeness and automated ontological classification.
5. **Graduate Programme Rules:** Extend the knowledge base with rules for master's and PhD student advising, thesis supervision tracking, and qualifying exam eligibility.
6. **Explainability Enhancements:** Add natural-language explanation templates for every rule, generating personalised advisory letters rather than structured reports.
7. **Persistence Layer:** Implement a SQLite or PostgreSQL database to store student sessions, track recommendation history, and enable longitudinal progress monitoring.

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